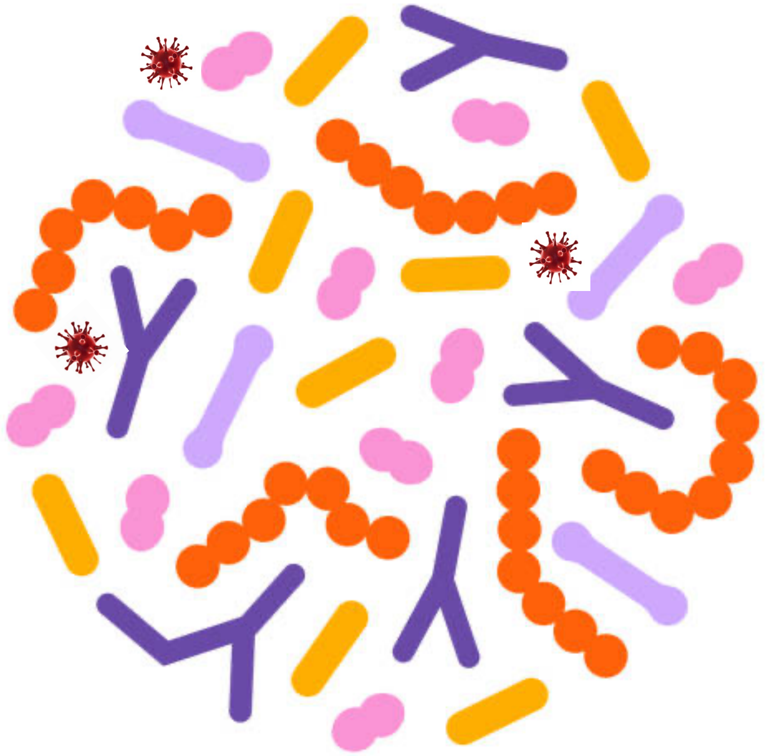


Introduction to Metagenomics

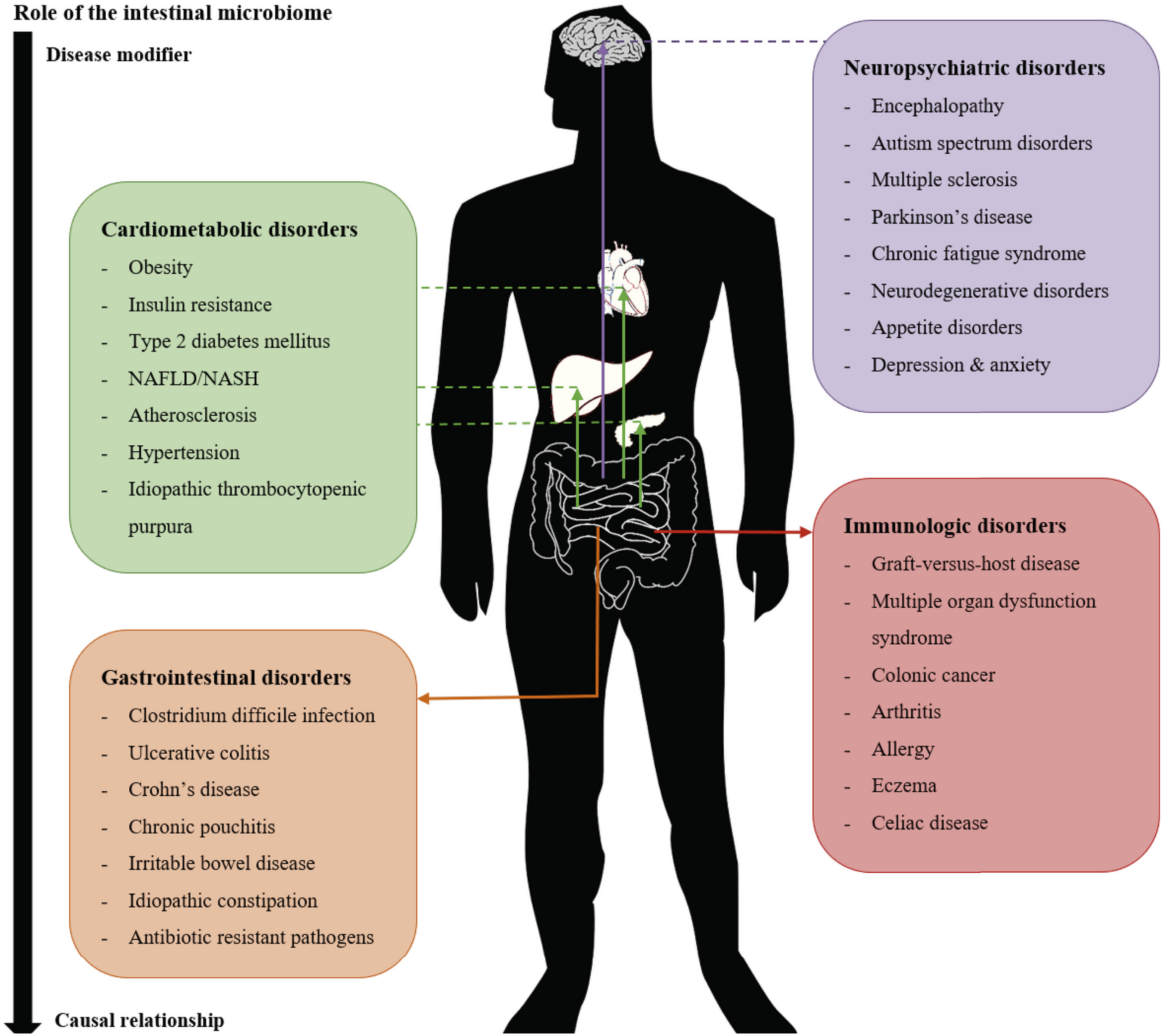
Data Science platform

DTU – Biosustain

Spring Data-type Courses

11th June 2025

Impact of microbiome studies in medical and planetary health



Microbiome has been associated with a variety of diseases

- Microbiota: cause or consequence of the disease?
- For some diseases it may also be a disease modifier or may metabolize the drugs administered to the diseased individuals

Obesity associated gut microbiome – a transmissible trait

An obesity-associated gut microbiome with increased capacity for energy harvest

Peter J. Turnbaugh, Ruth E. Ley, Michael A. Mahowald, Vincent Magrini, Elaine R. Mardis & Jeffrey I. Gordon

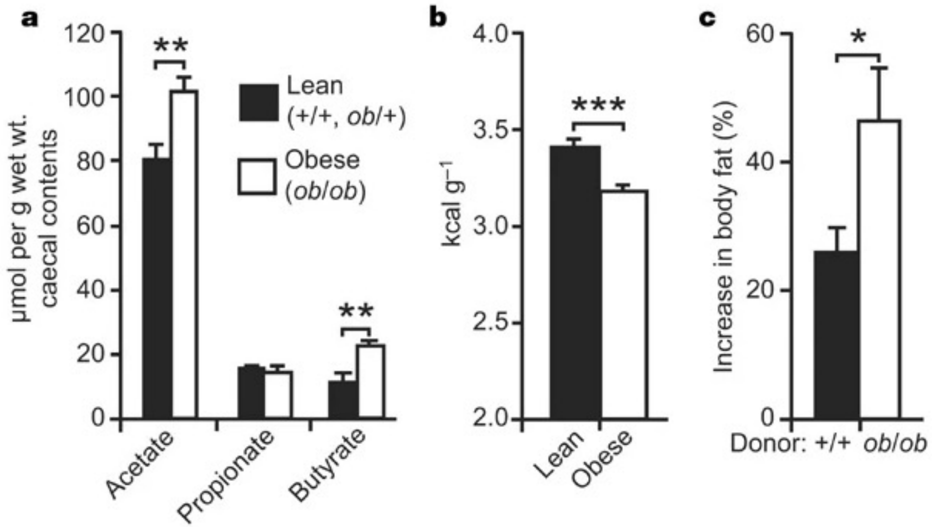
Nature **444**, 1027–1031 (2006) | Cite this article

https://doi.org/10.1038/nature05414

- Obese microbiome has an increased capacity to harvest energy from the diet

- **The trait is transmissible!**
  Colonization of germ-free mice with an 'obese microbiota' results in a significantly greater increase in total body fat than colonization with a 'lean microbiota'

Figure 3: Biochemical analysis and microbiota transplantation experiments confirm that the *ob/ob* microbiome has an increased capacity for dietary energy harvest.



Genetically obese mice had an increased concentration of the major fermentation end-products: butyrate and acetate

They also had significantly less energy remaining in their faeces

Germ-free mice colonized with an obese mice microbiota exhibited significantly greater increase in body fat

The Human Microbiome Project (HMP)

Article | Open access | Published: 13 June 2012

Structure, function and diversity of the healthy human microbiome

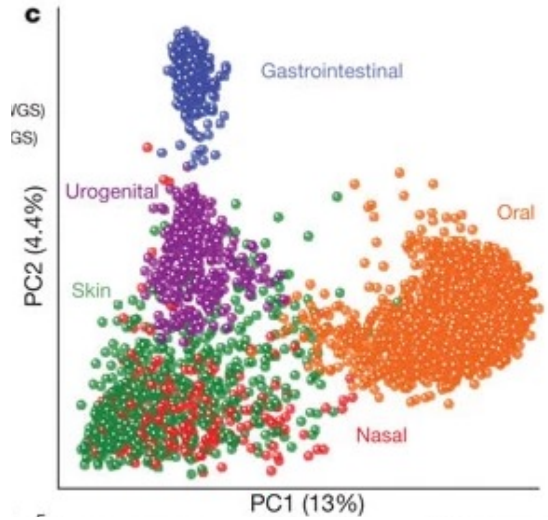
The Human Microbiome Project Consortium

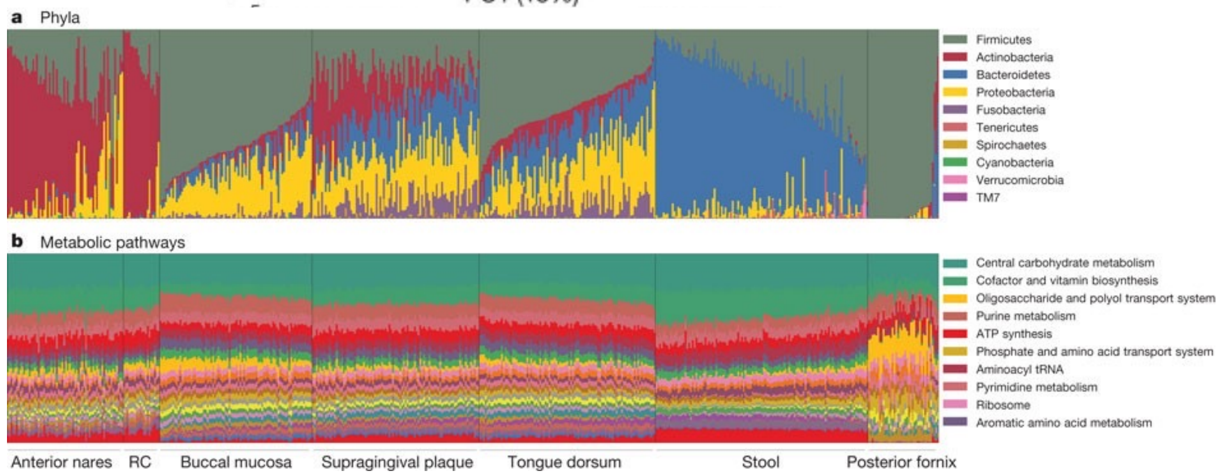
Nature **486**, 207–214 (2012) | Cite this article

450k Accesses | **1240** Altmetric | Metrics

https://doi.org/10.1038/nature11234

- Great initial resource to study the microbiome.
- Diversity of the human microbiome is concordant among measures, unique to each individual, and strongly determined by microbial habitat
- Carriage of microbial taxa varies while metabolic pathways remain stable within a healthy population





Richness of human gut microbiome correlates with metabolic markers (lessons of MetaHIT study)

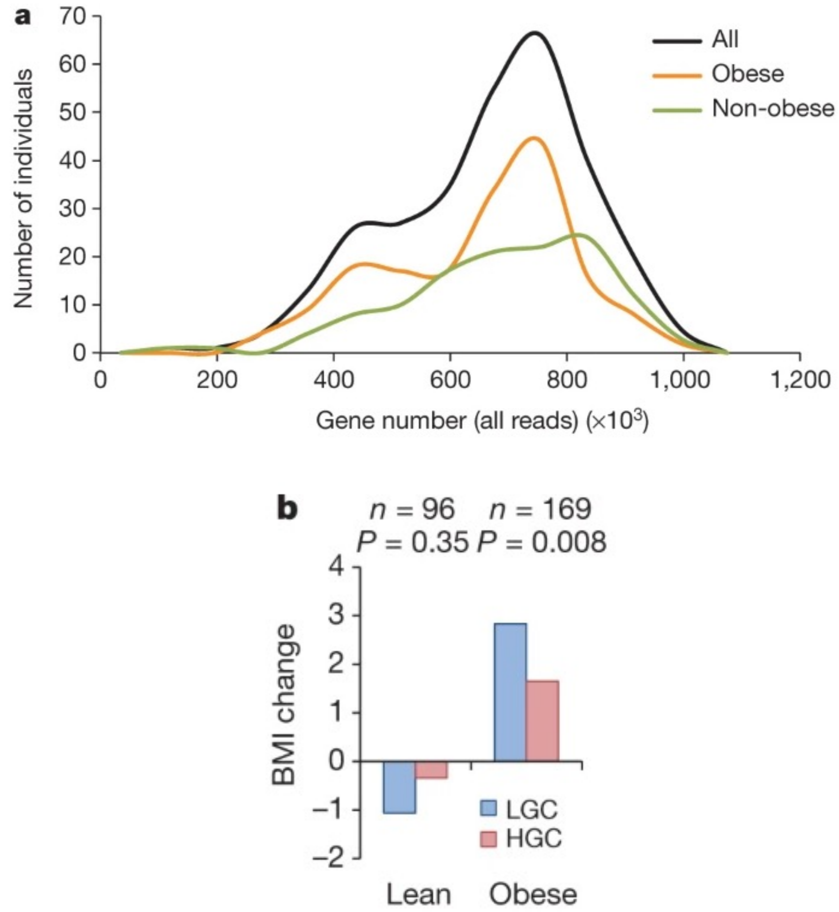
Article | Published: 28 August 2013

Richness of human gut microbiome correlates with metabolic markers

Emmanuelle Le Chatelier, Trine Nielsen, Junjie Qin, Edi Prifti, Falk Hildebrand, Gwen Falony, Mathieu Almeida, Manimozhiyan Arumugam, Jean-Michel Batto, Sean Kennedy, Pierre Leonard, Junhua Li, Kristoffer Burgdorf, Niels Grarup, Torben Jørgensen, Ivan Brandslund, Henrik Bjørn Nielsen, Agnieszka S. Juncker, Marcelo Bertalan, Florence Levenez, Nicolas Pons, Simon Rasmussen, Shinichi Sunagawa, Julien Tap, MetaHIT consortium, ... Oluf Pedersen + Show authors

Nature **500**, 541–546 (2013) | Cite this article 10.1038/nature12506

Figure 1: Distribution of low and high gene count individuals ($n = 292$).



- Individuals with a low bacterial richness are characterized by more marked overall adiposity, insulin resistance and dyslipidaemia and a more pronounced inflammatory phenotype
- Individuals with lower number of genes (gene richness) and higher BMI had less efficient weight loss when following a diet

Fecal material transplantation (FMT) as experimental cure for *Clostridioides difficile*


The NEW ENGLAND JOURNAL of MEDICINE

CURRENT ISSUE SPECIALTIES TOPICS

ORIGINAL ARTICLE

Duodenal Infusion of Donor Feces for Recurrent *Clostridium difficile*

Authors: Els van Nood, M.D., Anne Vrieze, M.D., Max Nieuwdorp, M.D., Ph.D., Susana Fuentes, Ph.D., Erwin G. Zoetendal, Ph.D., Willem M. de Vos, Ph.D., Caroline E. Visser, M.D., Ph.D., +5 , and Josbert J. Keller, M.D., Ph.D. Author Info & Affiliations

Published January 31, 2013 | N Engl J Med 2013;368:407-415 | DOI: 10.1056/NEJMoa1205037 | VOL. 368 NO. 5

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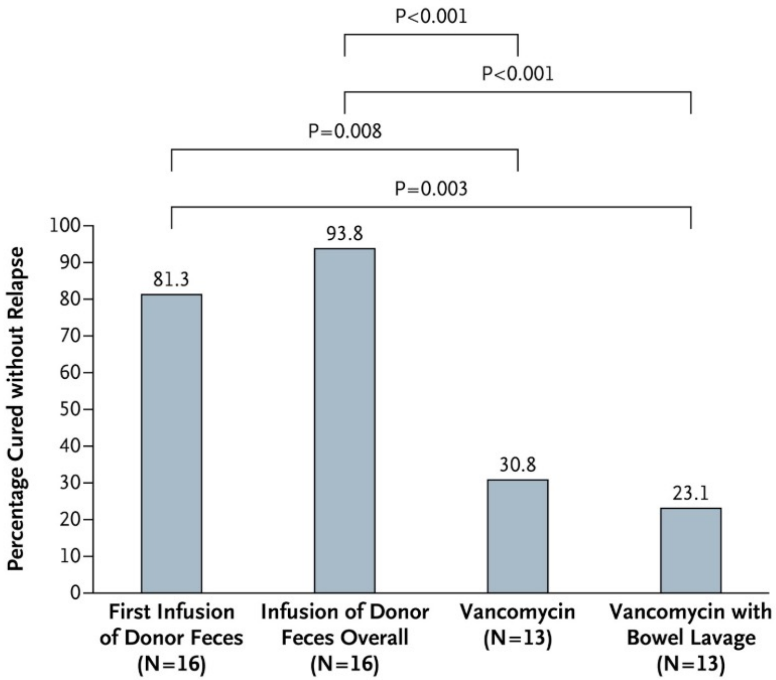
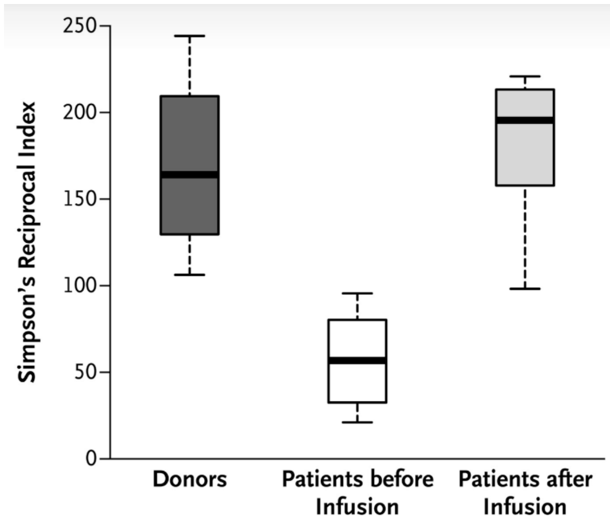
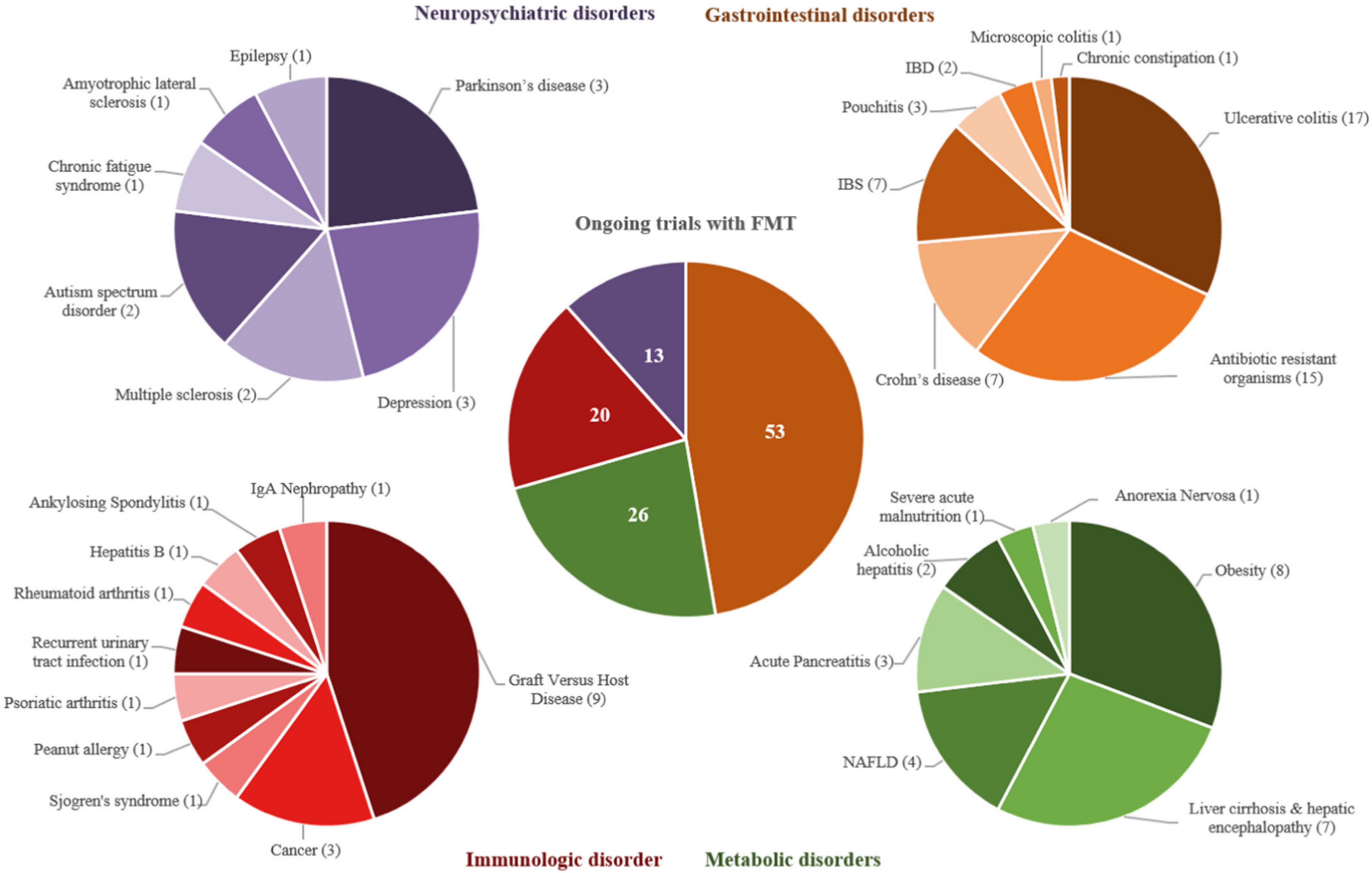
10.1056/NEJMoa1205037





FMT succesfully used to cure *Clostridioides difficile* infection with higher percentages of cured patients without relapse compared to the standard treatment (vancomycin)

Diversity (Simpson's index) of patients after FMT resemble more the donors diversity


Neuropsychiatric disorders
Epilepsy (1)
Parkinson's disease (3)
Amyotrophic lateral sclerosis (1)
Chronic fatigue syndrome (1)
Autism spectrum disorder (2)
Multiple sclerosis (2)
Depression (3)
Gastrointestinal disorders
Microscopic colitis (1)
IBD (2)
Chronic constipation (1)
Pouchitis (3)
Ulcerative colitis (17)
IBS (7)
Crohn's disease (7)
Antibiotic resistant organisms (15)
Ongoing trials with FMT
13
20
53
26
Ankylosing Spondylitis (1)
IgA Nephropathy (1)
Hepatitis B (1)
Rheumatoid arthritis (1)
Recurrent urinary tract infection (1)
Psoriatic arthritis (1)
Peanut allergy (1)
Sjogren's syndrome (1)
Cancer (3)
Graft Versus Host Disease (9)
Immunologic disorder
Severe acute malnutrition (1)
Anorexia Nervosa (1)
Alcoholic hepatitis (2)
Obesity (8)
Acute Pancreatitis (3)
NAFLD (4)
Liver cirrhosis & hepatic encephalopathy (7)
Metabolic disorders

Resident gut bacteria can affect patient responses to cancer immunotherapy

REPORT

Gut microbiome influences efficacy of PD-1–based immunotherapy against epithelial tumors

BERTRAND ROUTY, EMMANUELLE LE CHATELIER, LISA DEROSA, CONNIE P. M. DUONG, MARYAM TIDJANI ALOU, ROMAIN DAILLÈRE, AURÉLIE FLUCKIGER, MERIEM MESSAOUDENE, CONRAD RAUBER, [...], AND LAURENCE ZITVOGEL +38 authors Authors Info & Affiliations

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 91-97 • DOI: 10.1126/science.aan3706

DOI: 10.1126/science.aan3706

REPORT

The commensal microbiome is associated with anti–PD-1 efficacy in metastatic melanoma patients

VYARA MATSON, JESSICA FESSLER, RIYUE BAO, TARA CHONGSUWAT, YUANYUAN ZHA, MARIA-LUISA ALEGRE, JASON J. LUKE, AND THOMAS F. GAJEWSKI

Authors Info & Affiliations

SCIENCE • 5 Jan 2018 • Vol 359, Issue 6371 • pp. 104-108 • DOI: 10.1126/science.aao3290

DOI: 10.1126/science.aao3290

REPORT

Gut microbiome modulates response to anti–PD-1 immunotherapy in melanoma patients

V. GOPALAKRISHNAN, C. N. SPENCER, L. NEZI, A. REUBEN, M. C. ANDREWS, T. V. KARPINETS, P. A. PRIETO, D. VICENTE, K. HOFFMAN, [...], AND J. A. WARGO +60 authors Authors Info & Affiliations

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 97-103 • DOI: 10.1126/science.aan4236

DOI: 10.1126/science.aan4236

In a series of back-to-back papers in Science:

- Routy et al. showed that abx consumption --> poor response to cancer immunotherapy (immunotherapeutic PD1 blockade)
- Low levels of *Akkermansia muciniphila* in lung and kidney cancer patients → oral supplementation of this bacteria to abx-treated mice restored response to immunotherapy
- Matson *et al.* and Gopalakrishnan *et al.* studied melanoma patients receiving PD-1 blockade and found a greater abundance of “good” bacteria in the guts of responding patients.
- Non-responders had an imbalance in gut flora composition, which correlated with impaired immune cell activity.

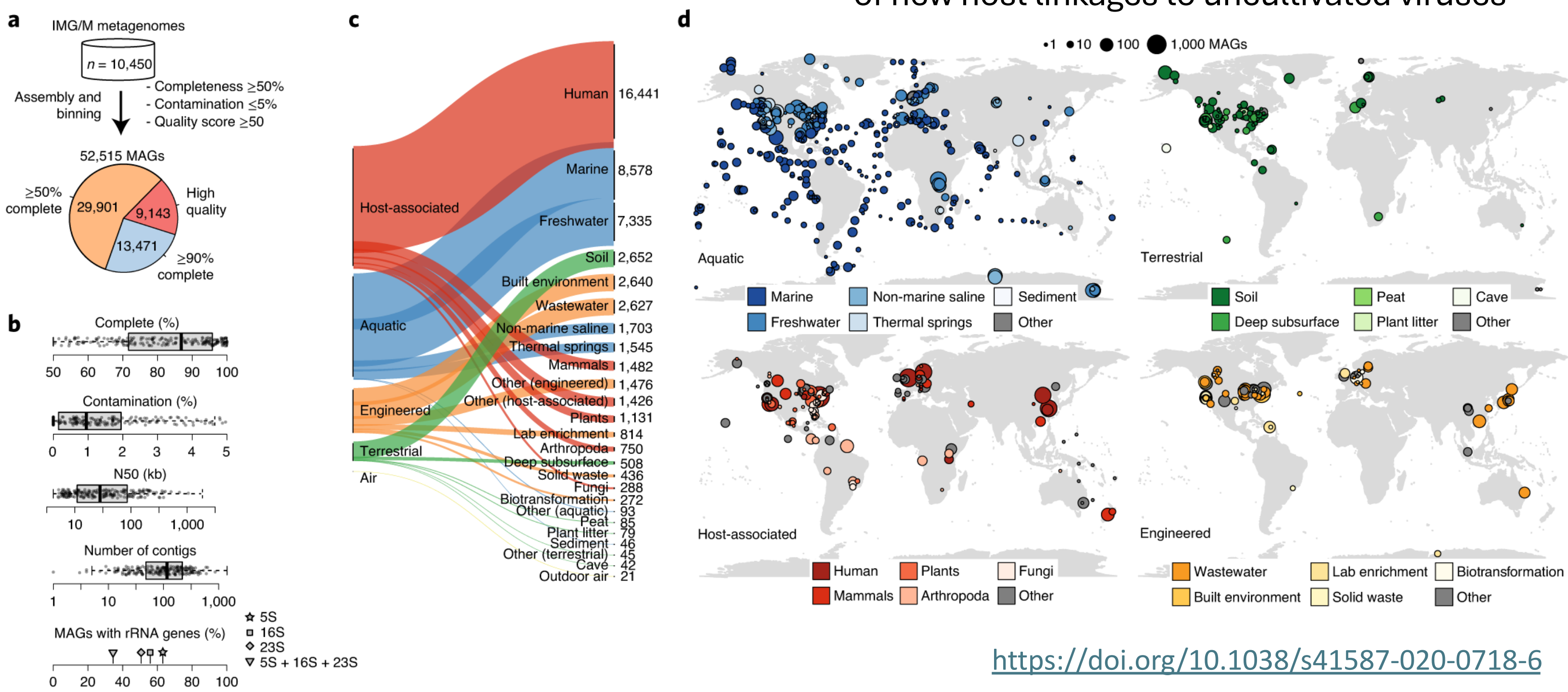
A genomic catalog of Earth’s microbiome

A genomic catalog of Earth’s microbiomes

Stephen Nayfach, Simon Roux, Rekha Seshadri, Daniel Udwary, Neha Varghese, Frederik Schulz, Dongying Wu, David Paez-Espino, I-Min Chen, Marcel Huntemann, Krishna Palaniappan, Joshua Ladau, Supratim Mukherjee, T. B. K. Reddy, Torben Nielsen, Edward Kirton, José P. Faria, Janaka N. Edirisinghe, Christopher S. Henry, Sean P. Jungbluth, Dylan Chivian, Paramvir Dehal, Elisha M. Wood-Charlson, Adam P. Arkin, IMG/M Data Consortium, … Emiley A. Eloe-Fadrosh + Show authors

Nature Biotechnology **39**, 499–509 (2021) | Cite this article

- >10,000 metagenomes collected from diverse habitats comprising host associated, aquatic, terrestrials, air and engineered microbiomes
- 52,515 metagenome-assembled genomes representing 12,556 novel candidate species from 135 different phyla
- Performing metabolic modelling, understanding secondary-metabolite biosynthetic potential and for resolving thousands of new host linkages to uncultivated viruses



https://doi.org/10.1038/s41587-020-0718-6

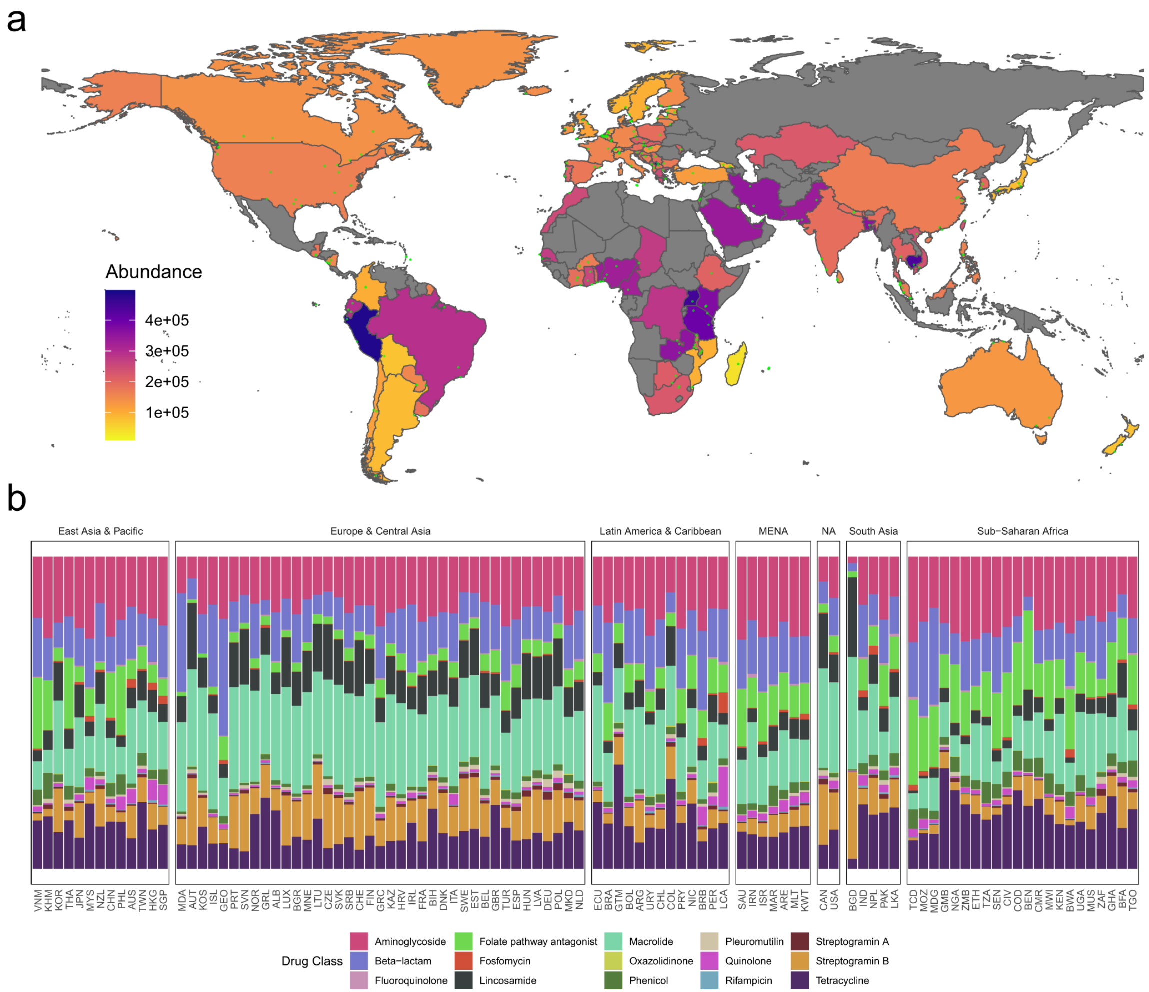
Antimicrobial resistance is a major threat to global health!

Genomic analysis of sewage from 101 countries reveals global landscape of antimicrobial resistance

Patrick Munk, Christian Brinch, Frederik Duus Møller, Thomas N. Petersen, Rene S. Hendriksen, Anne Mette Seyfarth, Jette S. Kjeldgaard, Christina Aaby Svendsen, Bram van Bunnik, Fanny Berglund, Global Sewage Surveillance Consortium, D. G. Joakim Larsson, Marion Koopmans, Mark Woolhouse & Frank M. Aarestrup

Nature Communications **13**, Article number: 7251 (2022) | Cite this article

https://doi.org/10.1038/s41467-022-34312-7



- The global resistome based on sewage-based monitoring
- Metagenomic samples revealed regional patterns reflecting taxonomical changes
- Understanding the emergence, evolution, and transmission of individual antibiotic resistance genes (ARGs) is essential to develop sustainable strategies to fight this threat

Microbial community at the Deepwater Horizon oil spill site

JOURNAL ARTICLE

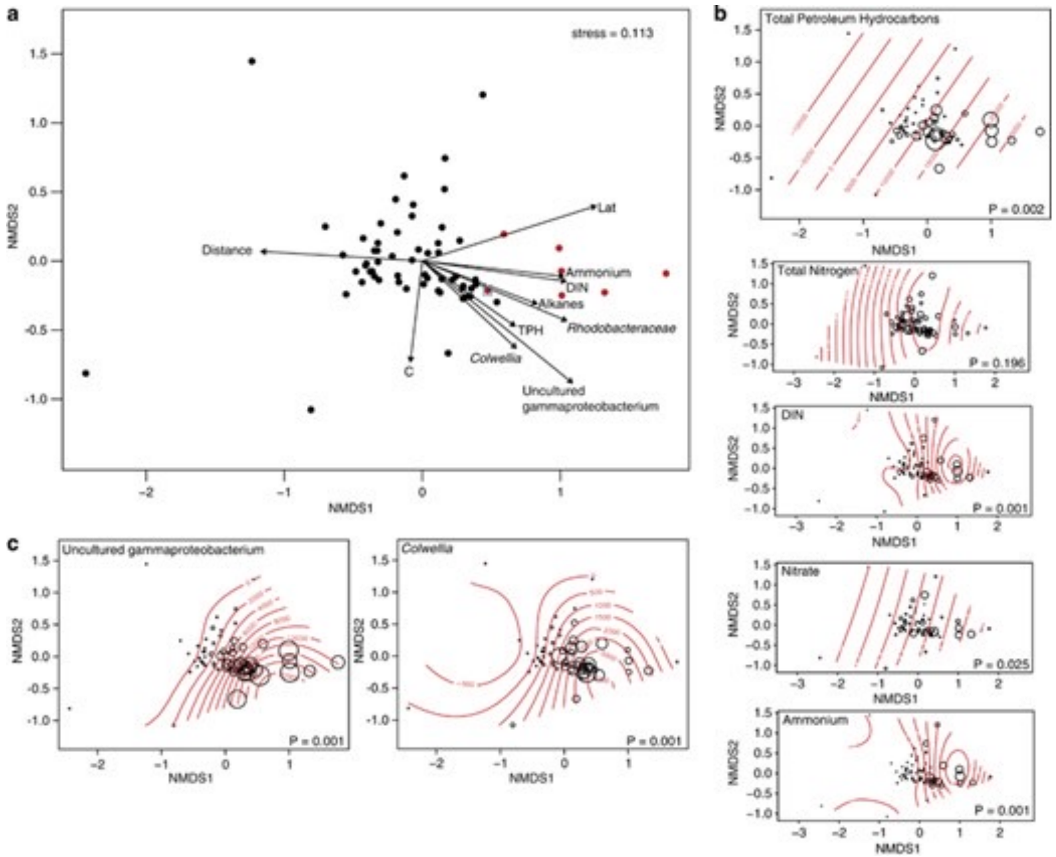
Metagenomics reveals sediment microbial community response to Deepwater Horizon oil spill

Olivia U Mason, Nicole M Scott, Antonio Gonzalez, Adam Robbins-Pianka, Jacob Bælum, Jeffrey Kimbrel, Nicholas J Bouskill, Emmanuel Prestat, Sharon Borglin, Dominique C Joyner ... Show more

The ISME Journal, Volume 8, Issue 7, July 2014, Pages 1464–1475,

https://doi.org/10.1038/ismej.2013.254

- Metagenomic Analysis of sediment samples from 64 sites in the Gulf of Mexico, characteryzing microbial community's response to the oil spill
- Uncultured *Gammaproteobacterium* and *Colwellia* species were enriched in oil-impacted sediments (hydrocarbon degradation)
- An increase in genes associated with hydrocarbon degradation pathways (aliphatic and aromatic hydrocarbon breakdown), indicating adaptive microbial response to oil contamination
- The study emphasized the importance of microbial communities in natural bioremediation processes.



Global ocean microbiome (Tara oceans)

Article | Open access | Published: 11 June 2018

Nitrogen-fixing populations of Planctomycetes and Proteobacteria are abundant in surface ocean metagenomes

Tom O. Delmont, Christopher Quince, Alon Shaiber, Özcan C. Esen, Sonny TM Lee, Michael S. Rappé, Sandra L. McLellan, Sebastian Lücker & A. Murat Eren

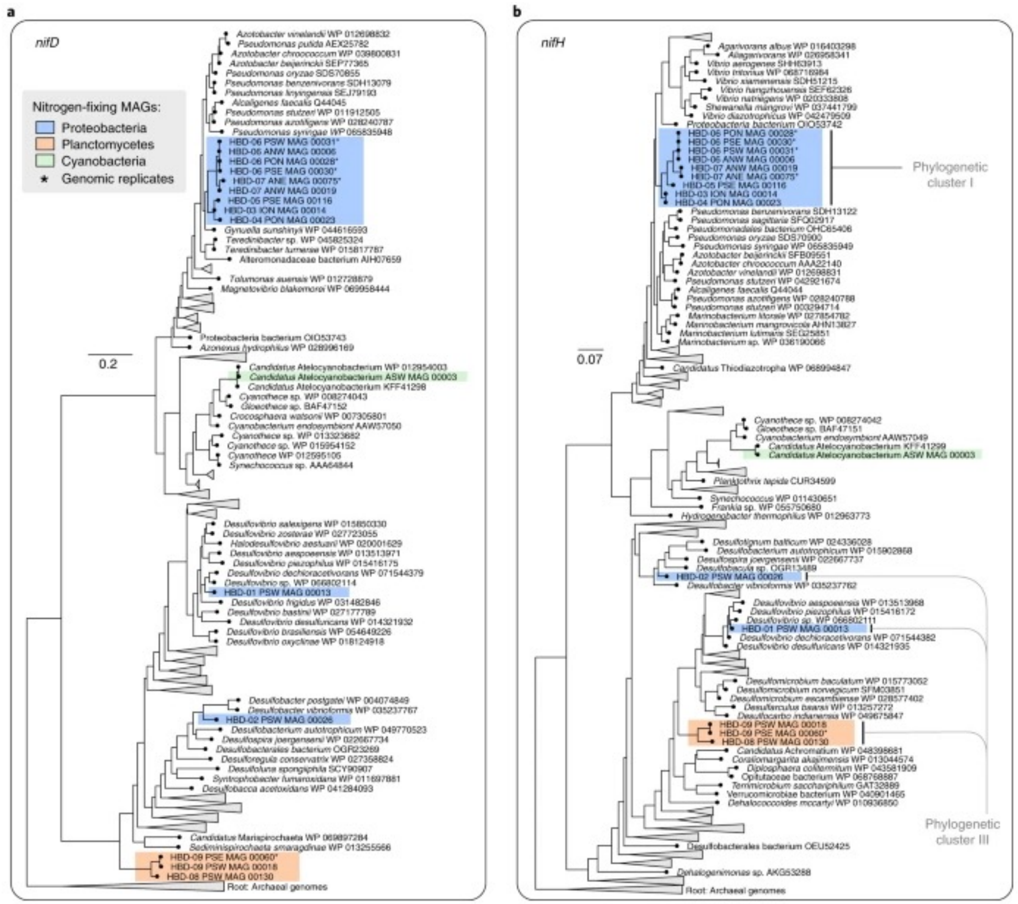
Nature Microbiology **3**, 804–813 (2018) | Cite this article

https://doi.org/10.1038/s41564-018-0176-9

- Characteryzed ~1,000 non-redundant microbial genomes from the TARA Oceans metagenomes

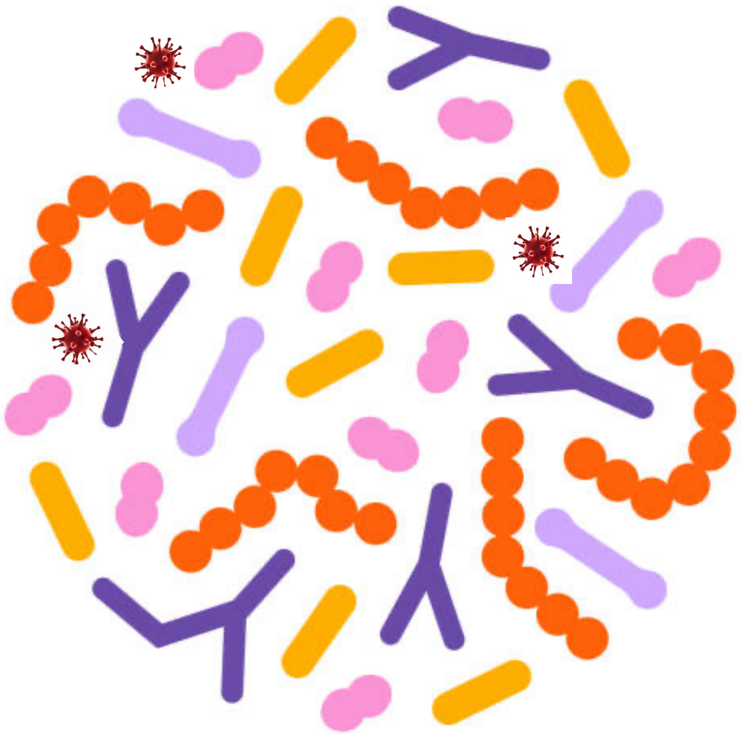
- Nitrogen fixation in the surface ocean impacts global marine nitrogen bioavailability and therefore the microbial primary productivity

- Non-cyanobacterial diazotrophs inhabiting surface waters of the open ocean, which correspond to lineages within the Proteobacteria and Planctomycetes



What is metagenomics?

Microbiome definition

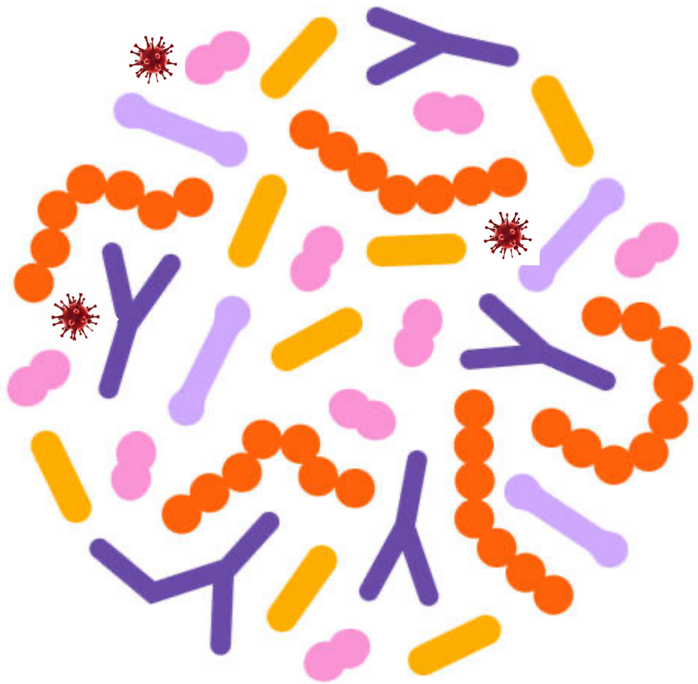


Microbiome definition

Microbiome refers to an microbial community in an specific environment

Inhabitants:
- Bacteria
- Archaea
- Fungi
- Viruses
- Protozoa
- …

Environment:
- Soil
- Ocean
- Wastewater plant
- Gut
- Skin
- …



Organism products:
- Genes
- Proteins
- Pathways
- Metabolites

In contrast to a monoculture (all cells belong to one organism), in a microbiome we have cells of many different organisms → microbial community

What a microbiome study usually answer?

What a microbiome study usually answer

Who is in the samples?

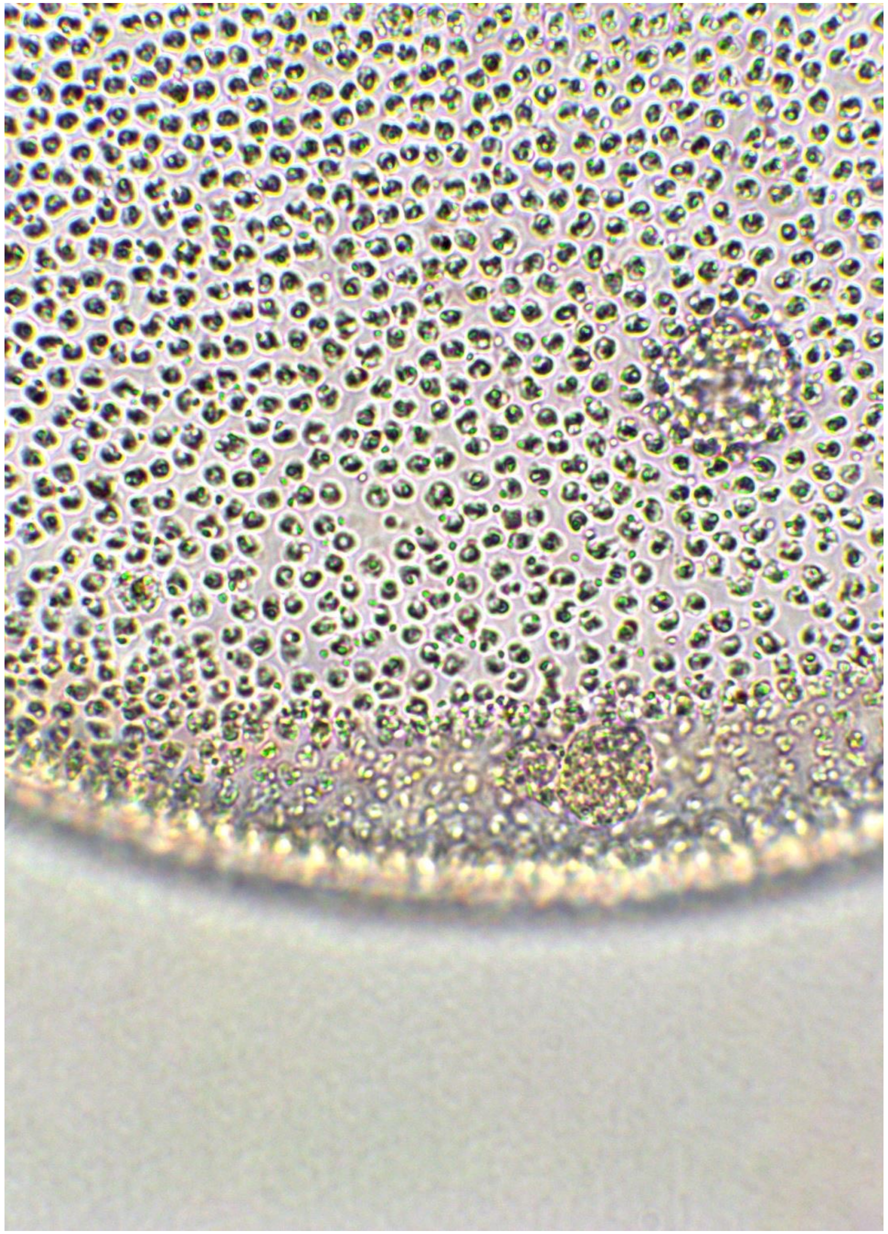
- What is the relative proportion of the different members
- What is the total abundace (number of cells) (by absolute abundace)
- How abundant are bacteria, archaea, protozoa, virus,…
- How diverse is the microbial community
- How evenly distributed is it?

What are these organisms doing?

- What functions potentially do they perform
- What substrates they consume?
- What metabolites they produce?

How do they perform their activities?

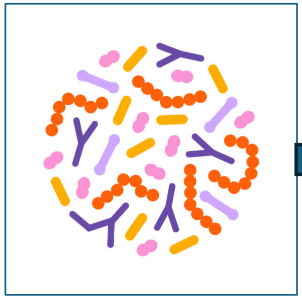
- Which genes and/or pathways are present?



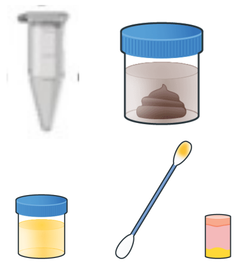
What is Metagenomics?

- Study of the genetic material sampled directly from the environment (fecal, skin, soil, marine,… samples)
- No need of culturing organisms – many of the organisms that we characterize can not even be cultured – we can study all the organisms in the microbial community
- Uses high-throughput DNA sequencing technology to analyze microbial diversity, taxonomic and functional profiles, and microbial interactions

Experimental design

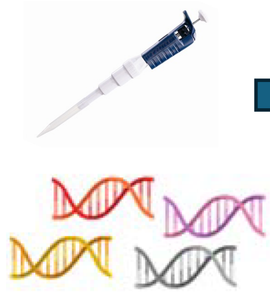


DNA Sampling
spatula, swabs, filtration units, soil cores,...



DNA Storage
freezing, preservation buffers,...

DNA Extraction
Kits to lyse the cells and release the DNA



DNA Sequencing
library prep: fragmentation, end repair, adapter ligation

Taxonomical and functional characterization of the sequenced reads (fastQ files)

Microbiome abundances

OTU	Sample 1	Sample 2	Sample 3
A	0.60	0	0.70
B	0.24	0.20	0.10
C	0.10	0	0
D	0.05	0	0
E	0.01	0	0
F	0	0.80	0.20
Total	1.0	1.0	1.0

Sample collection

- To obtain a **representative and uncontaminated snapshot of a microbial community** in a specific environment (e.g., gut, soil, water, skin,…)
- **Sterile tools** and containers need to be used
- Microbial DNA can degrade rapidly. A good preservation ensures the microbial community structure remains as close to in situ as possible.
    - **Freezing** (-20 to -80C) for long-term storage
    - Preservation buffers if freezing is delayed
- For sample transportation use **cold chain logistics** (ice packs, dry ice, liquid nitrogen)
- **Label** clearly with metadata: time, location, environmental conditions, etc.
    - Important for downstream analysis and reproducibility

DNA Extraction

- To obtain **high-quality, unbiased, and inhibitor-free DNA** from a complex matrix (e.g., soil, feces, water)
- Isolate **total microbial DNA** from all organisms in a sample (bacteria, archaea, fungi, viruses—depending on context)
- Steps:
  - **Cell Lysis:** The goal is to **lyse all types** of microbes, including tough ones like Gram-positive bacteria or spores.
    - Mechanical disruption (Bead beating)
    - Chemical/enzymatic lysis (detergent/enzyms (lysozyme, proteinase K, Rnase)

  Kits are preferred in metagenomics for consistency, ease, and reduced contamination risk (e.g., Qiagen, Zymo, MoBio/DNeasy)
  - **Contaminant Removal:** Use specialized kits with inhibitor removal steps (e.g., PowerSoil DNA Kit)
  - **DNA Purification:** remove proteins, RNA, lipids, and inhibitors
  - **Quantification & Quality Check**

Common DNA extraction kits:
- QIAGEN DNeasy PowerSoil Pro Kit
- QIAGEN QIAamp PowerFecal Pro DNA Kit
- ZymoBIOMICS DNA Miniprep Kit

Sequencing

- To read the **nucleotide sequences** of DNA fragments extracted from a microbial community — enabling identification of **who is present** and **what they can do**
- **Library preparation:** DNA must be **fragmented** and **tagged with adapters** for compatibility with sequencing platforms.
- Include negative and positive **controls** (e.g., mock communities)
- Choose **depth** wisely: deeper for complex samples or analysis (e.g., soil, strain resolution, single nucleotide variants(SNVs)), shallower for simpler ones
- Paired-end sequencing from Illumina (Sequencing by Synthesis):
    - https://www.youtube.com/watch?v=HMyCqWhwB8E

Platforms

Platform	Features	Read Length	Use Case
Illumina	High accuracy, short reads	~150–300 bp	Most common for shotgun & amplicon
Oxford Nanopore	Long reads, portable	up to 100 kb+	Assembly, rare taxa detection
PacBio HiFi	Long + accurate reads	~10–25 kb	Functional profiling, assembly
Ion Torrent	Medium throughput	~200–400 bp	Less common but used in clinical

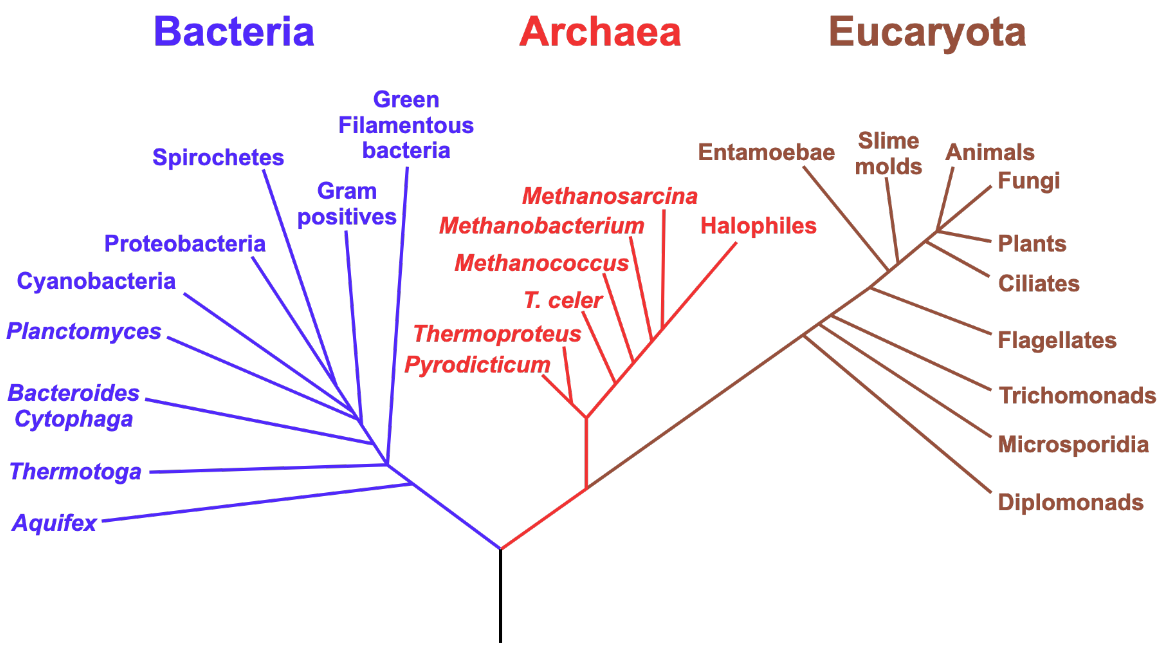
Taxonomical profiling

Taxonomical profiling

- Process of identifying and quantifying the microbial taxa present in a sample - this can be bacteria, archaea, viruses, fungi, ...
- Typically from metagenomic or amplicon sequencing data (metabarcoding).
- The most common taxonomical profiling methods can be grouped based on the type of sequencing data used:
  - Amplicon based profiling: 16S, 18S, ITS rRNA sequencing – e.g. 16S rRNA gene is used to profile bacteria and archaea
  - Whole-genome shotgun (WGS) metagenomics involves random sequencing of all genetic material in a sample:
    - Reference-based methods: direct mapping of reads to genes, proteins or genomes
    - Assembly-based methods

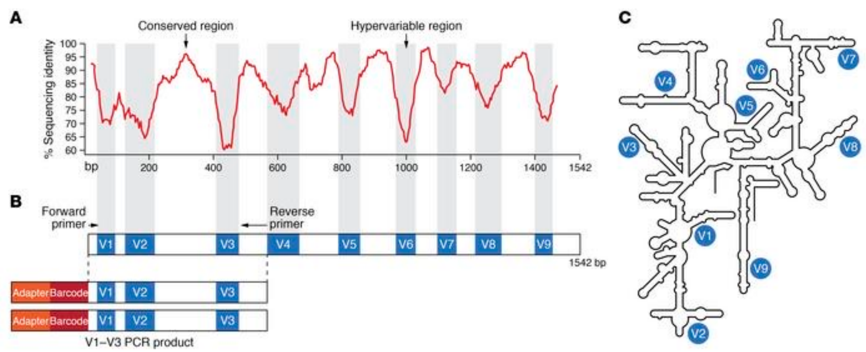
16S amplicon sequencing

Phylogenetic Tree of Life



Universal phylogenetic tree based on the 16S rRNA gene sequence comparisons. Pace, N. A molecular view of microbial diversity and the biosphere. Science276:734-740. 1997.

- 16S rRNA gene is an ancient gene that encodes the RNA component of the small (30S) subunit of the prokaryotic ribosome
- As it is conserved in many organisms, traditionally used to infer phylogenetic trees to figure out species evolution (Present also in mitochondria and chloroplasts of eukaryotes)
- 16s gene has
    - conserved regions: spread around all organisms and easy to amplify
    - variable unique regions: to tag specifically species



J Clin Invest. 2022;132(7):e154944. https://doi.org/10.1172/JCI154944.

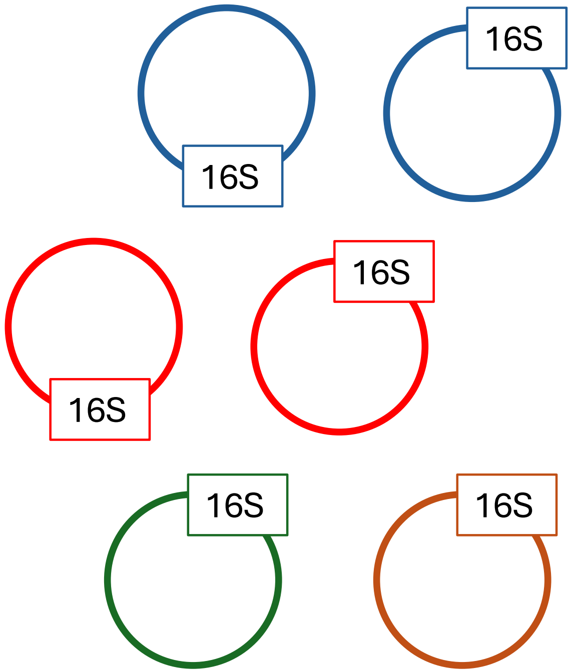
16S amplicon sequencing

DADA2
Amplicon Sequencing. Exactly.

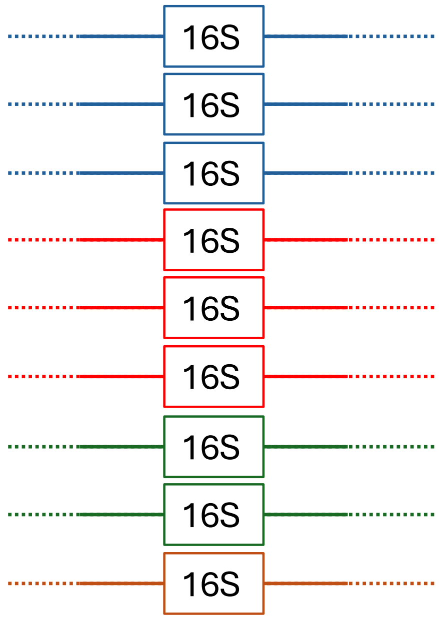
assign taxonomy directly to the amplicon sequence variants therefore allowing more precise taxonomic placement than OTUs

silva
high quality ribosomal RNA databases

green genes
16S rRNA gene database and workbench compatible with ARB
greengenes.lbl.gov



Microbial genomes in our sample carrying 16S rRNA gene

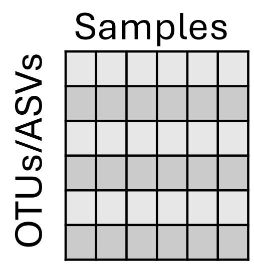


PCR amplification of 16S rRNA gene to produce a pool of DNA containing only 16S genes

T T **C** A A C C G A G A T A T G C
T T **C** A A C C G A G A T A T G C
T T **C** A A C C G A G A T A T G C
T T T A A C C G A **T** A T A T G C
T T T A A C C G A **T** A T A T G C
T T T A A C C G A **T** A T A T G C
T T T A A C **A** G A G A T A T G C
T T T A A C **A** G A G A T A T G C
T T T A A C C G A G A T A T **A** C

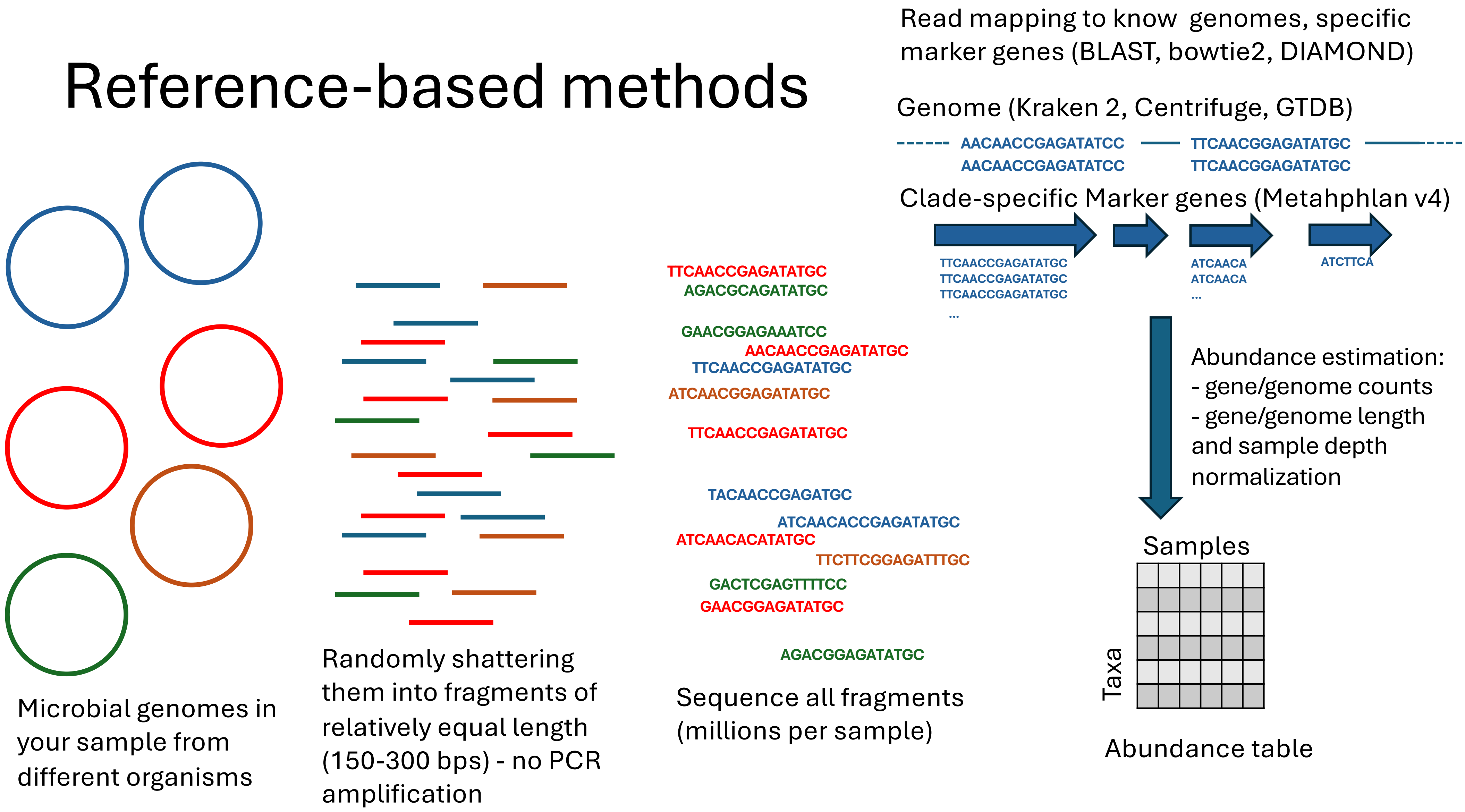
Sequencing all fragments

Bin by operational taxonomic units and compare to known 16S databases



Abundance table of operational taxonomic units (OTUs) or amplicon sequence variants (ASVs) – resolution up to Genus

Reference-based methods



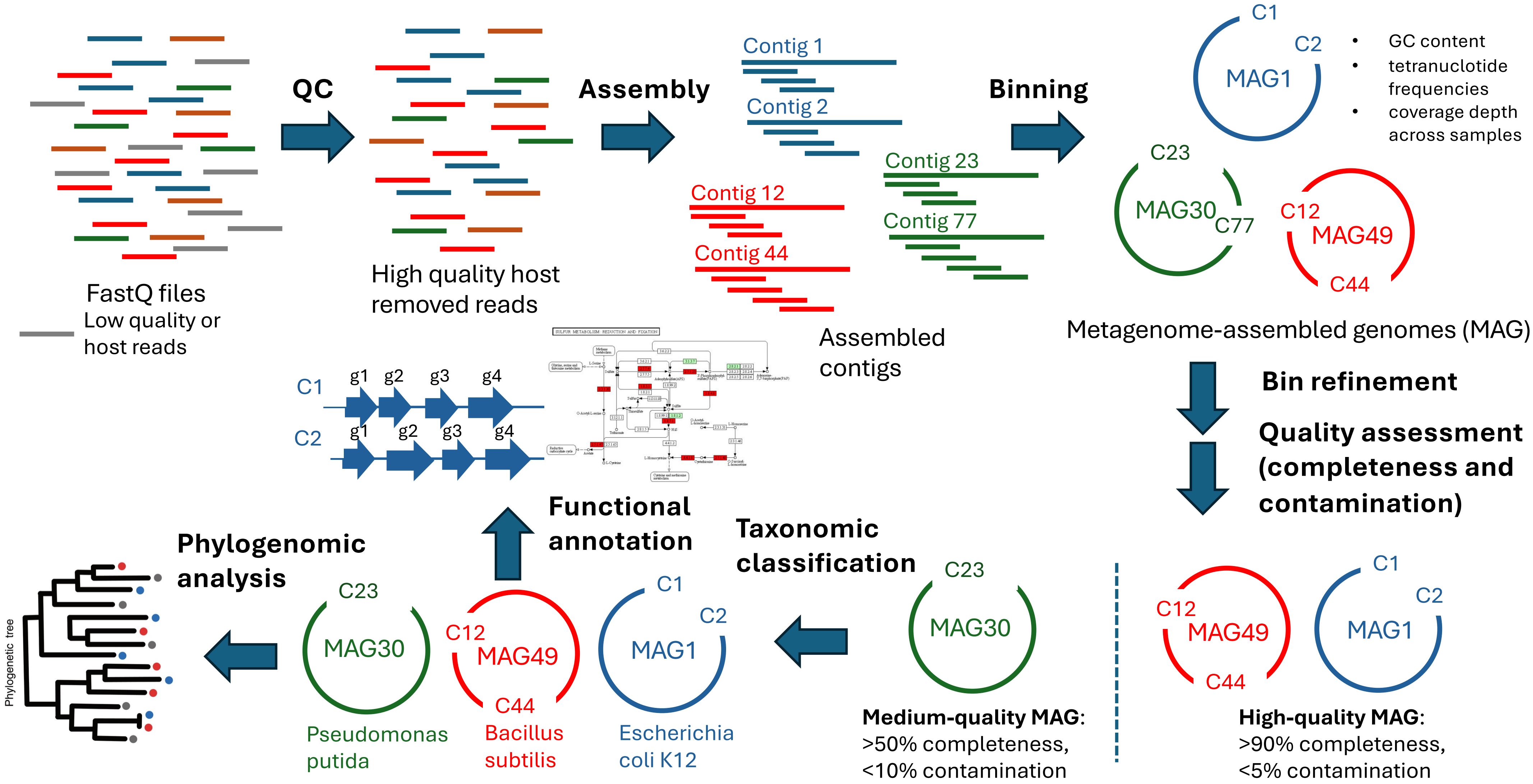
Amplicon and read-based shotgun metagenomics tools

Method	Tools	Databases
16S Amplicon-based Profiling	**QIIME 2**: A popular pipeline supporting DADA2, Deblur, and multiple classifiers (Naive Bayes, etc.). **DADA2**: High-resolution method that infers amplicon sequence variants (ASVs). **USEARCH / VSEARCH**: Clustering-based OTU picking methods. **Mothur**: OTU-based method using different clustering and alignment strategies. **RDP Classifier**: Naive Bayesian classifier for assigning taxonomy.	**SILVA** **Greengenes** (deprecated but still used) **RDP** **UNITE** (for fungi)
Shotgun Metagenomics-based Reference-based profiling	**MetaPhlAn4**: Uses clade-specific marker genes for fast, accurate profiling. **Kraken2 / KrakenUniq**: K-mer based, ultra-fast classification against a database of genomes. **Centrifuge**: K-mer based, designed for large reference databases and low-memory use. **HUMAnN / HUMAnN3**: Often used alongside MetaPhlAn for functional and taxonomic profiling. **StrainPhlAn**: Extension of MetaPhlAn for strain-level profiling.	**ChocoPhlAn** (used by MetaPhlAn) **RefSeq / GenBank** (used by Kraken/Centrifuge) **GTDB** (Genome-based taxonomic reference)

Choice depending on:

- **Speed vs. Accuracy**: Kraken2 is fast; MetaPhlAn is more accurate for species-level identification.
- **Resolution**: DADA2 and MetaPhlAn can provide high-resolution profiling.
- **Data Type**: Amplicon (16S) vs. shotgun metagenomics.
- **Resource Requirements**: Kraken needs large memory for big databases; MetaPhlAn is lighter.

Assembly-based methods: Metagenome-assembled genomes (MAG)



MAG construction tools

https://pubmed.ncbi.nlm.nih.gov/28898207/

Step	Goal	Tools	Notes
Quality Control	Remove low-quality reads, adapters, and contaminants (e.g. host reads)	fastp, adapterremoval, Trimmomatic, BBduk	Output: Clean reads
Assembly	Assemble short reads into longer contigs.	MEGAHIT, metaSPAdes	Output: Assembled contigs
Binning	Group contigs that belong to the same genome	MetaBAT2, MaxBin2, CONCOCT, VAMB	Output: bins of contigs belonging to the same genome Using coverage profiles and sequence composition (GC content, tetranucleotide frequency).
Bin refinment	Improve bin quality and merge results from different binning tools	DASTool, metaWRAP	Output: Refined bins
Quality Assessment	Assess completeness and contamination of bins (MAGs)	CheckM, BUSCO	Metrics usually used: High-quality MAG: >90% completeness, <5% contamination. Medium-quality MAG: >50% completeness, <10% contamination.
Taxonomic classification	Assign taxonomy to MAGs	GTDB-Tk, Kraken2, CAT/BAT	Output: taxonomically annotated MAGs
Functional annotation	Predict genes and metabolic functions	Prokka, EggNOG-mapper	Out put: functionally annotated MAGs
Phylogenomic analysis	Compare MAGs evolutionarily	PhyloPhlAn, Anvi’o, IQ-TREE	Output: Phylogenetic trees based on MAGs

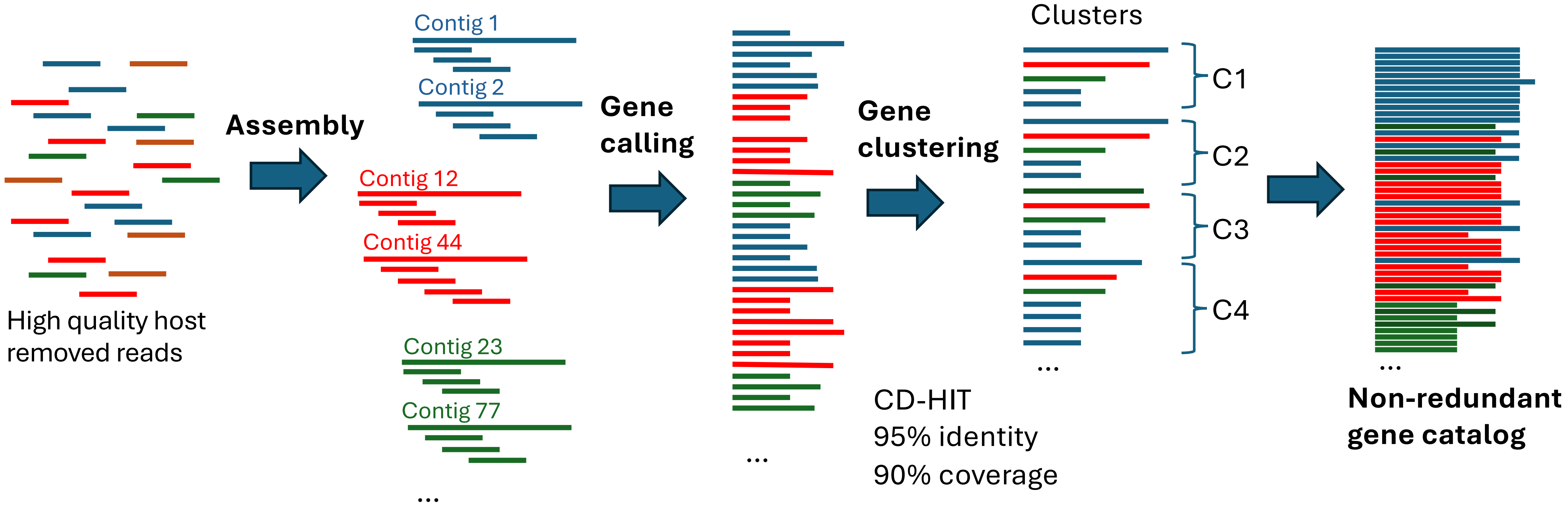
Comparing the three methods

	Amplicon/16S sequencing	Shotgun metagenomics	
		Reference - based	Assembly - based
Pros	• Can handle low biomass / non-bacterial contamination • Rough taxonomy of novel species • Less expensive	• High resolution (subspecies, genes SNVs) • Can detect all DNA in the sample (also virus, fungus, protozoa,...) • Fast and scales well • Finds rare entities	• High resolution (subspecies, genes SNVs) • Can detect all DNA in the sample (also virus, fungus, protozoa,...) • Can identify novel genomes, genes and synteny patterns
Cons	• Low resolution • Limited to bacteria or archaea • No subspecies or strains resolution • Biases due to PCR amplification / amplicon copy number variation	• Does not work well for low biomass • More expensive than 16S • Limited to what is known in the databases	• Does not work well for low biomass • More expensive than 16S • Misses rare entities • Computationally very intensive

Functional profiling

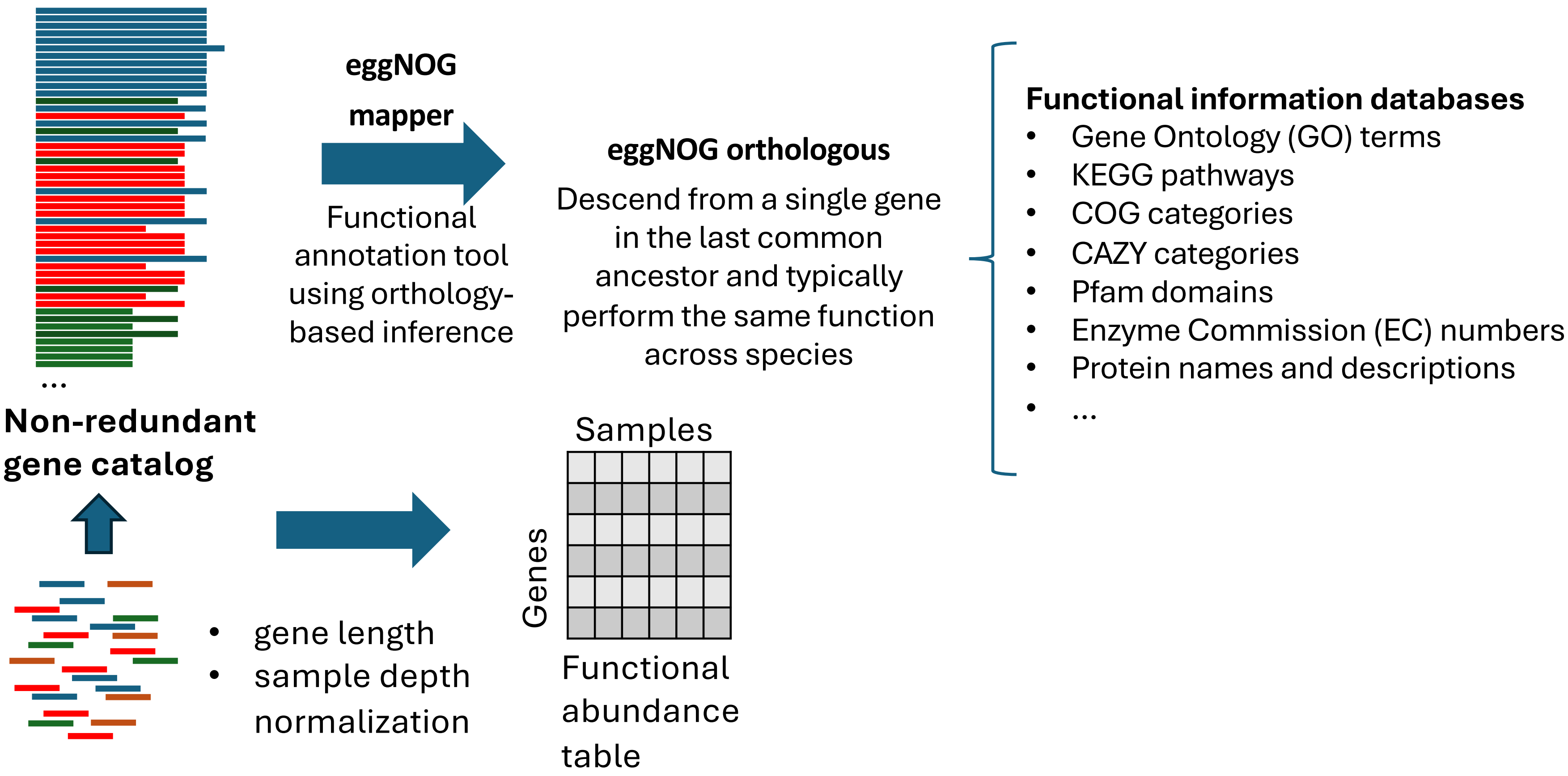
Building a microbial gene catalog

To do functional annotation you can always map reads to genes (read/reference-based annotation)
A common resource used in functional annotation is building a gene catalog
The beginning of the process is the same as the assembly method



Annotating the gene catalog and getting functional abundance

Reads can be mapped to the catalog genes to infer gene abudance and those genes can be mapped to several functional information databases (e.g KEGG database, PATRIC Database, CARD or Resfams antibiotic resistance gene databases,...)



Why building a gene catalog?

- Annotation of genes across all metagenome (not only the known ones – novel or uncharacterized functions - new biochemical functions, pathways, or resistance mechanisms)

- Facilitates comparative and longitudinal studies (compare between environments or timepoints)
- Improves Annotation Efficiency (non-redundant)

- Reusable resource (valuable resource for ongoing large scale projects)

- Enhanced sensitivity in rare function detection (genes from low abundance taxa carrying potentially important functions (e.g ARGs, secondary metabolites)

Microbiome data

Microbiome data

- **Sparsity**
- **Dinamic range**
- Usually expressed as Relative, not Absolute abundance
- Compositional data

Relative abundace (%)

Area	1	2	3	1	2	3	1	2	3
Condition	X	X	X	Y	Y	Y	Z	Z	Z
Relative abundance Taxa or functional feature	10	11	20	12	14	16	25	34	22
	70	49	56	0.9	0.5	0	0	0.1	0
	0	0	0	0.4	0.1	0	0	0	0
	0.3	0.4	0.45	0	0.1	1	2	3	4
	2	2	3	0	0	0.01	1	1	2

Samples

Microbiome data

- Sparsity
- Dinamic range
- **Usually expressed as Relative, not Absolute abundance**
- **Compositional data**

Microbiome data is compositional:

- Sampling of a fraction of the total genomic DNA
- Different sequencing depth between samples
- Both sampling depth and sequencing depth can vary several orders of magnitude between samples
- Absolute abundances are normally not known – abundances are relative and constraint to sum to 1 (or 100%)
- Proportions or percentages of total reads per sample — not true cell counts.
- Relative abundances are susceptible to compositional effects

https://pubmed.ncbi.nlm.nih.gov/27306058/

Compositional data

Consider read counts as a proxy for cell abundance (not perfect as 16S gene can have multiple gene copies in some genomes, and in shotgun metagenomics we have different genome sizes – larger genomes contributing with more reads per cell, extraction efficiency, library prep bias, etc ...)

	S1 cells	S1 Rel ab %	S2 cells	S2 Rel ab %
Taxa 1	250	25	250	33.33
Taxa 2	250	25	250	33.33
Taxa 3	300	30	50	6.67
Taxa 4	200	20	200	26.67
Total	1000	100	750	100

- Only Taxon 3 reduced its abundance after antibiotic treament, but 1,2, and 4 seem that they increase
- Affects downstream analysis and interpretations
    - e.g. correlations or regression assume independent variables
    - analysis alternatives and data transformations have been proposed: center-log-ration transformation (clr), ALDEx2, ANCOM
    - however, they introduce practical limitations — especially with zero handling, computational cost, and difficult interpretation

Ideal scenario: absolute abundances

If we want to know **true cell counts**, we need to:

- Add **internal spike-in controls** (known amounts of reference DNA).
- Use **qPCR or flow cytometry** to estimate **total bacterial load**.
- This allows you to correct read counts to reflect **absolute abundance**, not just proportions.

Diversity measures

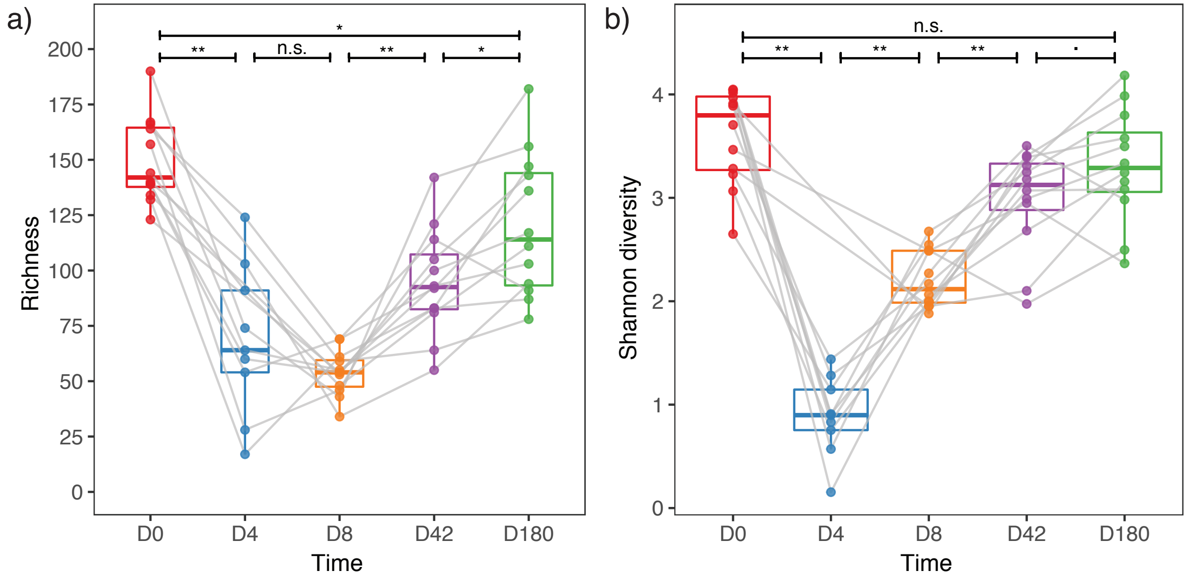
Diversity measures: Alpha diversity

- Diversity within a sample
  - **Richness**: number of unique taxa found in a sample (species, OTUs, ASVs,...)
  - **Shannon Index**: Accounts for both richness and evenness (how balanced the community is)
    - **Higher values** = More diverse and evenly distributed community
  - Other indices: Simpson, Chao1, Faith's PD (for phylogenetic diversity)

- Shannon index:

$$H' = -\sum_{i=1}^{S} p_i \cdot \ln(p_i)$$

  - S = total number of species
  - pi = proportion of species *i* in the sample

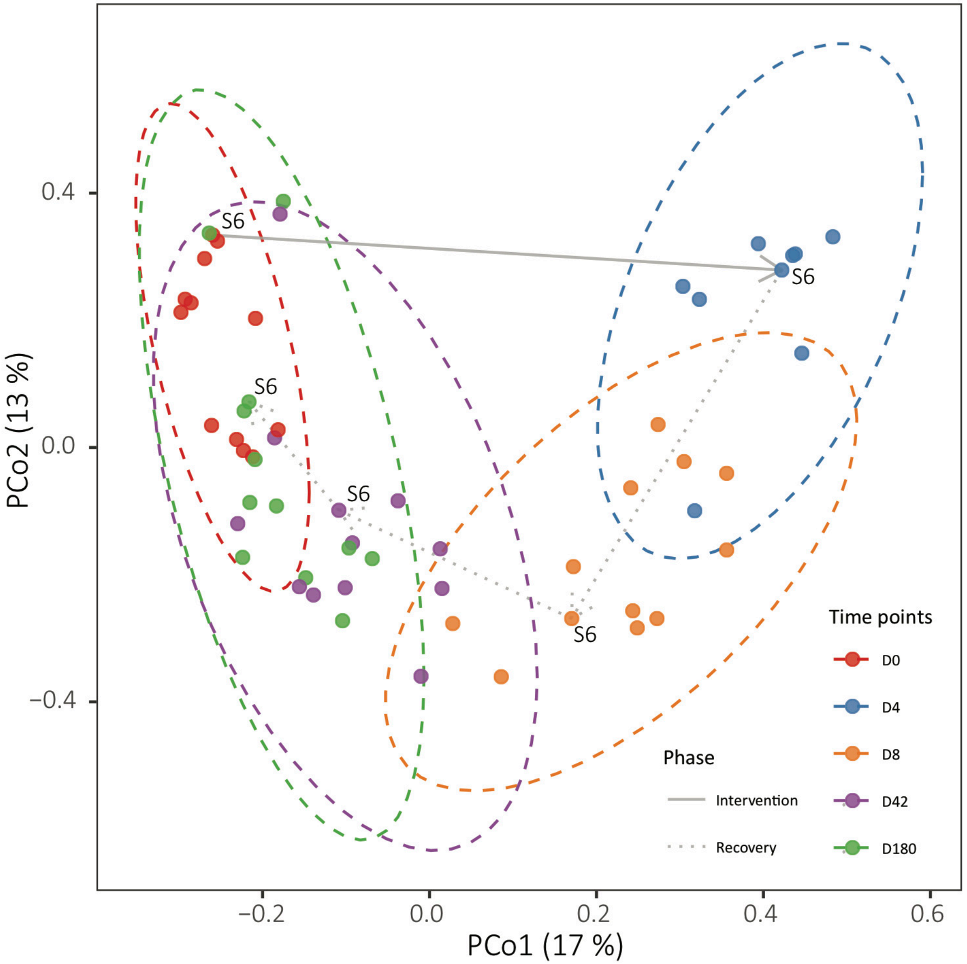


https://pubmed.ncbi.nlm.nih.gov/30349083/

Diversity measures: Beta diversity

- Diversity between samples (community composition dissimilarity)
- For example:
  - **Bray-Curtis dissimilarity** (presence/absence + abundance-weighted)
  - **Jaccard distance** (presence/absence)
  - **UniFrac** (phylogenetic distances: weighted/unweighted)

- These distances between samples are calculated based on the abundance and projected on the two first dimensions of a PCoA.
- other ordination methods are also used: NMDS, tSNE or simply PCA,...)



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Challenges and limitations

Challenges and limitations of metagenomics studies

- Shotgun metagenomics have higher cost than 16S amplicon but you get higher taxonomical resolution and functional information
- Sampling biases can occur when collecting environmental samples which can also lead to misrepresentation of the microbial community
- It is also challenging to extract microbes DNA from low biomass samples - highly dependant on the sampling method and the its accuracy
- Microbial genomes available and deposited in reference databases are biased towards model organisms, pathogens and cultivable bacteria
- Metagenomic sequencing can detect the presence of microorganisms but cannot easily determine if they are pathogenic (do they have their toxins? Are they active?)

https://pubmed.ncbi.nlm.nih.gov/28898207/

Challenges and limitations of metagenomics studies

- Functional profiling is hindered by the lack of annotation for many genes
- "Microbial dark matter" – microbes not yet characterized can be recovered by assemble-based methods, but still many reads are not used in the assembly – sequencing noise, DNA contamination, and bacteria and plasmids that remain obscure even after some parts of their genomes are assembled
- "Live or dead" dilemma – DNA persists in the environment after the death of the host cell, sequencing results may not be representative of the active microbial population
- Snapshot of the microbiome - many metagenomics studies are cross-sectional and can not assess intra- vs inter-subject variation – more longitudinal studies are needed
- Disentangling the cause from the effect – follow ups, longitudinal and prospective studies needed
- Metagenomics have not yet reached the level of standardization of other high-throughput techniques

https://pubmed.ncbi.nlm.nih.gov/28898207/